CS 229 Final Project: Yoga Pose Classification with MediaPipe Pose Detection and Convolutional Neural Networks (CNNs)

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Introduction

- **Yoga pose classification** is useful for fitness apps, virtual coaching, and teaching yoga [1]
- It's challenging because many poses differ only in subtle details (ex: extended side angle vs. triangle = one bent knee)
- This project compares two approaches: **extracting body landmarks with MediaPipe** and training classifiers on geometric features [2, 3], versus **fine-tuning CNNs** directly on images
- Evaluated on **Yoga-82 dataset**
- Input: yoga images → Output: pose class (6/20/82 classes)
- **SVM** with geometric features (84.2%) rivals **ResNet-50** (85.3%) and outperforms **ResNet-18** (78.9%), showing structured features are competitive with deep learning at lower computational cost

Data

- Yoga-82 dataset consists of URLs for colored yoga pose images collected from the web, each labeled with pose names organized in a 3-level hierarchy (6, 20, and 82 classes)
- After filtering for accessible URLs and successful MediaPipe landmark detection, the final dataset contains 10,976 training and 4,195 test images across 82 pose classes, with moderate class imbalance

Reverse Warrior Pose (82)
Standing (6) Side-bend (20)

Bound Angle Pose (82)
Sitting (6) Legs in front (20)

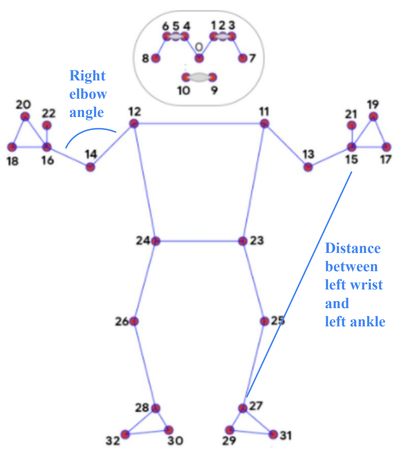
Locust Pose(82)
Reclining (6)
Downward-facing (20)

Revolved Head to Knee Pose (82)
Reclining (6) Up-facing (20)

Features

- MediaPipe Pose extracts 33 body landmarks per image, each with (x, y, z, visibility) coordinates:
- 33 * 4 = **132-dim features**

- I engineered 32 additional features: 10 joint angles (elbows, shoulders, hips, knees), 10 pairwise distances between key body parts, and 12 left-right symmetry measures:
- 132 + 32 = **164 features**



MediaPipe Feature Extraction and feature engineering [7]

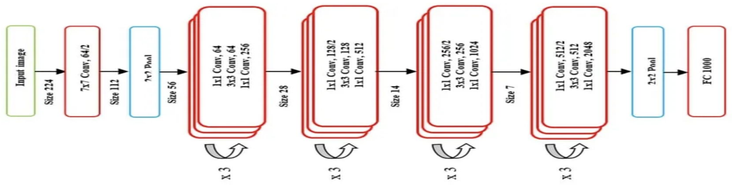
Models

Random Forest [5]: An ensemble of 200 decision trees, each trained on a bootstrap sample with random feature subsets. Final predictions are made by majority vote across all trees.
Support Vector Machine (SVM) [6]**:** Finds the maximum-margin decision boundary using an RBF kernel to capture nonlinear boundaries between poses. The kernel equation:

$$K(x, x') = e^{-\gamma\|x-x'\|^2} \quad \gamma\text{: kernel width}$$

K-Nearest Neighbors (KNN) [5] **:** Classifies each sample by majority vote among its k=7 nearest neighbors, weighted by inverse distance.
Multi-layer Perceptron (MLP) [5]**:** A neural network with three hidden layers (256, 128, 64) and ReLU activations, outputting class probabilities via softmax function
ResNet-18 & ResNet-50 [4]**:** Pretrained CNNs fine-tuned on raw images, using residual connections y = F(x) + x. All layers frozen except layer 4; final classifier replaced with dropout and a new fully connected layer to output 6, 20, or 82 classes.

- **ResNet-18:** 18 layers with residual blocks (each with 2 conv layers) ~11M params
- **ResNet-50:** 50 layers with bottleneck residual blocks (each with 3 conv layers) ~25M params
  - Deeper architecture allows richer feature representation for pose distinctions



ResNet-50 architecture [8]

Results

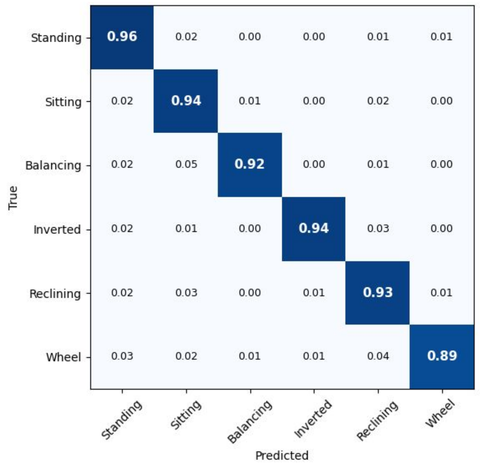
- Combined features (164-dim) performed better than body landmarks alone (132-dim) across all classifiers and class levels, with the largest improvement at the 82-class level in test accuracies

Test Accuracies

Model	Features	6-class	20-class	82-class
KNN	combined	90.0%	85.6%	77.1%
Random Forest	combined	91.1%	87.3%	80.9%
MLP	combined	92.4%	89.1%	83.8%
SVM	**combined**	**93.6%**	**90.6%**	**84.2%**
ResNet-18	images	89.8%	84.8%	78.9%
ResNet-50	**images**	**93.0%**	**89.9%**	**85.3%**

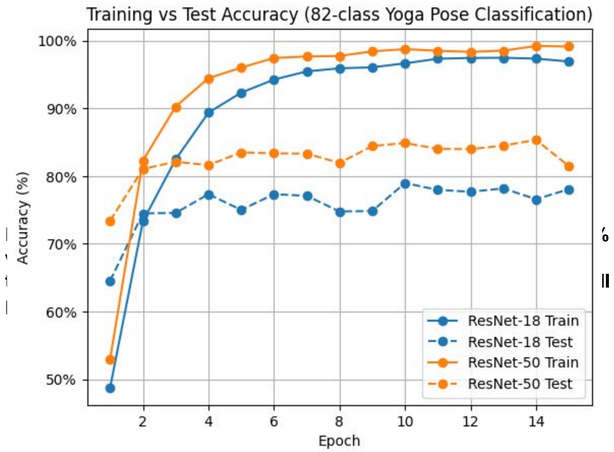
- SVM with combined features outperforms ResNet-18 and approaches ResNet-50
- This shows that geometric features are competitive with deep learning
- The deeper ResNet-50 performs better than ResNet-18, with the highest improvement at the 82-class level

SVM 6-class Confusion Matrix (combined)



- SVM achieves strong per-class accuracy at the 6-class level, with most errors between poses with similar body configurations (ex: balancing ↔ sitting)

Overfitting Analysis



- Both CNNs show overfitting, with training accuracy reaching ~99% while test accuracy plateaus early
- ResNet-50 generalizes better than ResNet-18 (85.3% vs. 78.9%), but both are limited by the small per-class training set (~134 images)

Discussion

- Landmark-based models performed well, with SVM using combined geometric features achieving accuracy close to the best CNN
- SVM performed particularly well because the geometric landmark features likely create nonlinear relationships between poses, which the RBF kernel can capture
- Adding engineered geometric features improved performance across all classical classifiers, especially for the more difficult 82-class task
- ResNet-50 achieved the highest accuracy overall, likely because its deeper architecture can learn richer visual features from images

Future

- Explore hybrid models that combine geometric landmark features with image-based CNN features to leverage both structured pose information and visual appearance
- Investigate methods to reduce CNN overfitting, such as stronger data augmentation, more data or regularization

References
[1] Verma, M., Kumawat, S., Nakashima, Y., & Raman, S. "Yoga-82: A New Dataset for Fine-Grained Classification of Human Poses." IEEE/CVF CVPR Workshops, 2020.
[2] Bazarevsky, V., Grishchenko, I., Ravi, K., Kartynnik, T., & Grundmann, M. "BlazePose: On-device Real-time Body Pose Tracking." CVPR Workshop on CV for AR/VR, 2020.
[3] Google. "MediaPipe Pose Landmarker." https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker, 2023.
[4] He, K., Zhang, X., Ren, S., & Sun, J. "Deep Residual Learning for Image Recognition." IEEE CVPR, 2016.
[5] Pedregosa, F. et al. "Scikit-learn: Machine Learning in Python." JMLR, 12:2825–2830, 2011.
[6] Ng, A. & Ma, T. "CS229 Lecture Notes." https://cs229.stanford.edu/notes2022fall/main_notes.pdf, 2022.
[7] Google-ai-edge. "MediaPipe Pose." https://github.com/google-ai-edge/mediapipe/blob/master/docs/solutions/pose.md
[8] Mukherjee, S. "The Annotated ResNet-50." Towards Data Science, 2022. https://towardsdatascience.com/the-annotated-resnet-50-a6c536034758